Tactix public GF(3) Gaussian batch action
Certified reasoning checkpoint

Development report for Andrew

4 October 2026

Outcome

The chained GF(3) Gaussian reasoning path now reaches the public controller boundary. Commit `cd956aea` registers X14 `Gf3GaussianConsequence` at append-only rule ID 37, disabled reference StepId 26, and C parameter tag 40 in API version 6. Explicit programs and an opt-in random-controller producer can select it through the shared executor; DefaultPolicy remains disabled pending performance and regression acceptance.

This is a correctness and architecture checkpoint. No new solver benchmark, fixed-44 recovery run, WV/WV2 run, or regression campaign was launched. The saved `d170b4a5` comparison was not rerun.

Why the action is one complete batch

Gaussian elimination may derive several root units from the same authenticated closure. Applying one unit closes the action's assignment-free boundary, so a per-unit public action could make the remaining simultaneous deductions unreachable. Discovery therefore offers exactly one deterministic `g3b1:` identity for the complete closure.

That identity commits every supported unit or conflict, the complete `Source`/`Scale`/`Add` derivation, authentication references, and exact source clauses. Apply recaptures the live formula, rebuilds the algebraic and Boolean certificate, exact-matches the identity, and atomically publishes the whole batch. This preserves a controller-visible choice without hiding which subset the frontend selected.

Boundary	Contract	State
Offer	Mutation-free construction of one authenticated complete-batch key	pass
Apply	Fresh rebuild and exact identity match before any publication	pass
Proof	Ordinary RUP units/conflict in DSR, DRAT, or no-proof mode	pass
Replay	Typed C++ serialization and additive C v6 deep-copy	pass
Default scheduling	Explicit disabled row; production admission unmeasured	pending
Target recovery	Fixed-44 and WV/WV2 evidence	pending

Safety, proof, and accounting

Corrupt, stale, cross-kernel, interrupted, exhausted, or unsupported-proof requests decline before solver, proof, journal, or session mutation. One Kernel-bound session retains durable duplicate keys. Identity construction and total retained session identity bytes have separate limits and raw counters, avoiding an unmetered replay payload.

Only entailed unit clauses or the empty clause enter search. Intermediate case-split clauses remain proof-only. The rule adds no search auxiliary, RAT definition, PB step, or reconstruction entry. DSR, DRAT, and no-proof are supported; LRAT and PBP decline before mutation. Nonunit reduced rows remain outside this first public action.

The action journal records typed limits, exact premises, authenticated equations, reduced rows, original-source combinations, proof events, installed clauses, and action identities. Feedback separately records capture, bridge, certificate, validation, revalidation, proof, journal, identity, session, unit/conflict, and terminal-status measurements. Fuzz reach is keyed to the `gf3-gauss` attempt counter, which moves only when an exact batch enters the shared executor.

Validation

Independent review confirmed that offer is mutation-free and apply performs a fresh exact rebuild before atomic commit. The current Debug build passed 7/7 selected integration tests: GF(3) live/public, API instances, all online actions, inprocessing program, random online selection, and scheduling. The public test covers multi-unit publication, corrupt and stale identities, cross-kernel sessions, both byte limits, typed replay, shared online execution, random reach, C v6 validation, and DSR output.

ASan/UBSan passed the live and public tests 2/2 with leak detection disabled because LSAN is unsupported under the environment's tracing. The feedback inventory has 699 native fields and no wall timers. Effort classification, catalog/inventory, calculus/HTML synchronization, document size, progress, and Python syntax checks pass.

Campaign assessment and next work

The partial recovery campaign remains correct but incomplete. Six direct structured MDP instances were recovered quickly with independently checked models. `cabp` took about 1,026 seconds versus the accepted 222 seconds; Break timed out at 1,200 seconds and Timetable at 600 seconds. No wrong verdict or checker failure occurred, but partial phase guidance was never offered or consumed. The new X14 boundary has not yet been measured on those rows.

The next public vertical is Tseitin connected-component selection: expose an exact component payload, session, replay, feedback, and scheduling without a private frontend choosing the opportunity. Once benchmark work is authorized, run the three trajectory controls and targeted owner subsets, then the fixed 44 and the B28+C5, Rank1–50, WV100, and WV2–89 gates. Effective default/random fuzz must require X14 reach. Independent proof/model checking, the 1,800-second proof-check cap and slowdown rule, and N1–N4 remain mandatory.

The detailed record is `benchmarks/reports/2026-10-04-public-gf3-gaussian-action.md`.